

Credit_Risk_Project

Graduation Project Report

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1. Problem Statement

Banks receive a large number of loan applications and must decide which applicants are likely to repay their loans. Reviewing each application manually is slow and can be inconsistent from one reviewer to another. The goal of this project is to build a Machine Learning model that predicts whether a loan applicant is likely to default, so banks can make faster and more consistent lending decisions.

2. Dataset Description

The project uses the `credit_risk_dataset.csv` dataset, which contains 32,581 loan applicant records. Each record includes information about the applicant and the loan, such as:

- Person age, income, and employment length
- Home ownership status
- Loan intent, loan grade, loan amount, and interest rate
- Loan percent of income and credit history length
- Loan status (target variable: whether the applicant defaulted)

3. Methodology

The project followed a standard Machine Learning workflow, carried out in three phases:

- Data Cleaning: Removed unrealistic values (e.g. applicant age above 100, employment length above 80 years) and filled missing values in employment length and interest rate using the median.
- Feature Engineering: Converted categorical columns (such as home ownership and loan intent) into numeric form using one-hot encoding, and scaled numeric features using `StandardScaler`.
- Model Training: Split the data into 80% training and 20% testing sets, then trained five different classification models to compare their performance.

4. Results

The five models were evaluated on the test set using accuracy as the primary metric. The results are summarized below:

Model	Accuracy
Logistic Regression	86.5%
K-Nearest Neighbors (k=5)	89.2%
Random Forest	93.0%
XGBoost	93.1%
ANN (MLPClassifier)	91.6%

Random Forest and XGBoost achieved the highest accuracy (around 93%), outperforming the simpler Logistic Regression baseline. Because accuracy alone can hide important mistakes, a Confusion Matrix was also generated for the Random Forest model to check how many applicants were correctly vs. incorrectly classified. This showed that the model correctly identified most low-risk applicants, with a smaller number of misclassifications on the higher-risk group — an area that could be improved with further tuning.

5. Conclusion

This project successfully built and compared multiple Machine Learning models for credit risk prediction. Random Forest and XGBoost were identified as the best-performing models, reaching approximately 93% accuracy. The resulting model can help banks automate part of the loan approval process, reducing manual review time and supporting more consistent lending decisions. Future improvements could include testing additional models, handling class imbalance more explicitly, and deploying the model as a real-time scoring API.